DDL – Create & Alter

1. **Create a table `hr_dept_backup` as a full copy of `hr.departments`.**

```
SELECT *
FROM HR_DEPT_BACKUP
```

2. **Add a column `notes` VARCHAR2(100) to `hr_emp_backup`**

```
ALTER TABLE HR_DEPT_BACKUP ADD NOTES VARCHAR(20)

SELECT * FROM HR_DEPT_BACKUP
```

3. **Create a table `emp_50` from employees in department 50 only (all columns).**

```
CREATE TABLE EMP_50 AS SELECT * FROM HR.EMPLOYEES WHERE DEPAR
TMENT_ID='50'

SELECT * FROM EMP_50
```

4. **Add column `updated_at` DATE DEFAULT SYSDATE to your backup table**

```
ALTER TABLE HR_DEPT_BACKUP ADD UPDATE_AT DATE DEFAULT SYSDATE

SELECT * FROM HR_DEPT_BACKUP
```

5. **Create a table `dept_names` with only `department_id` and `department_name` from hr.departments.**

```
CREATE TABLE DEPT_NAMES AS SELECT DEPARTMENT_ID,DEPARTMENT_NA
ME FROM HR.DEPARTMENTS

SELECT * FROM DEPT_NAMES
```

6. **Modify column `notes` in hr_emp_backup to VARCHAR2(500).**

```
ALTER TABLE HR_DEPT_BACKUP MODIFY NOTES VARCHAR(500)

DESC HR_DEPT_BACKUP
```

7. **Create an empty table `emp_structure` with the same structure as hr.employees (no rows).**

```
CREATE TABLE EMP_STRUCT AS SELECT * FROM HR.EMPLOYEES WHERE 1
=0

SELECT * FROM EMP_STRUCT
```

8. **Rename table `hr_Dept_backup` to `hr_employees_archive`.**

```
RENAME HR_DEPT_BACKUP TO HR_EMP_ARCHIEVE

SELECT *
FROM HR_EMP_ARCHIEVE
```

9. **Add two columns to a backup table: `created_by` VARCHAR2(50) and `created_date` DATE.**

```
ALTER TABLE  EMP_STRUCT ADD (CREATED_BY VARCHAR(50),CREATED D
ATE);

SELECT * FROM EMP_STRUCT
```

10. **Create table `high_earners` from hr.employees where salary > 10000 (all columns).**

```
CREATE TABLE HIGHHER_EARNER AS SELECT * FROM HR.EMPLOYEES WHE
RE SALARY>'10000'
```

```
SELECT * FROM HIGHHER_EARNER;
```

11. **Drop the column `notes` from your backup table.**

```
ALTER TABLE EMPLOYEE DROP COLUMN name;
select * from Employee;
```

12. **M12. Create table `emp_salary_dept` with only employee_id, salary, department_id from hr.employees.**

```
CREATE TABLE EMP_SAL_DEPT (
  empId int,
  SALARY INT,
  dept_ID VARCHAR(10)
);

SELECT * FROM EMP_SAL_DEPT
```

13. **Truncate the table `emp_50` (or whatever copy table you created).**

```
TRUNCATE TABLE EMP_BACK;

SELECT * FROM HR.EMPLOYEES
```

14. **Rename column `remarks` to `comments` in your backup table.**

```
alter table bonus rename column comm to comments;

select * from bonus;
```

15. **Create a table `dept_emp_count` with department_id and a literal 0 as column `emp_count` (one row per department).**

```
create table emp_dept as select LOC,0 as emp_count from dept;

SELECT * FROM EMP_DEPT
```

16. **Add column `status` VARCHAR2(20) DEFAULT 'ACTIVE' to a backup table**

```

 ALTER TABLE EMP_DEPT ADD STATUS VARCHAR(20) DEFAULT'ACTIVE';

SELECT * FROM EMP_DEPT
```

17. **Create table `emp_hire_2005` from hr.employees where EXTRACT(YEAR FROM hire_date) = 2005.**

```
CREATE TABLE EMP_HIRE_2005 AS SELECT * FROM EMP WHERE EXTRACT
(YEAR FROM HIREDATE)=1981;

SELECT * FROM EMP_HIRE_2005;
```

18. **Modify column `status` to VARCHAR2(30).**

```
ALTER TABLE EMP_DEPT MODIFY STATUS VARCHAR(30);
-- FOR SIZE CHANGE
DESC EMP_DEPT
```

19. **Create an empty table with the same structure as hr.departments. Name it `dept_template`.**

```
CREATE TABLE DEPT_TEMPLATE AS SELECT * FROM EMP WHERE 1=0;
-- WON'T DISPLAY ROWS
DESC DEPT_TEMPLATE
```

20. **Add column `audit_id` NUMBER(10) to your backup table.**

```
 ALTER TABLE DEPT_TEMPLATE ADD AUDIT_ID NUMBER(10);

--Table altered.

SQL> DESC DEPT_TEMPLATE
```

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1. **Create a table `emp_dept_summary` that has one row per department with columns department_id, department_name (from hr.departments), and a computed column total_sal (use a subquery or join to get SUM(salary) per department).**

```
CREATE TABLE EMP_DEPT_SUMMARY AS SELECT SUM(SALARY) AS TOTAL_
SAL,DEPARTMENT_ID
FROM HR.EMPLOYEES GROUP BY DEPARTMENT_ID;

SELECT * FROM HR.EMPLOYEES;
```

2. **Create table `emp_backup_80` from hr.employees for department 80, but only columns employee_id, first_name, last_name, salary, commission_pct.**

```
CREATE TABLE EMP_BACK_80
AS SELECT EMPLOYEE_ID,FIRST_NAME,LAST_NAME,SALARY,
COMMISSION_PCT,DEPARTMENT_ID
FROM HR.EMPLOYEES
WHERE DEPARTMENT_ID='80';

SELECT * FROM EMP_BACK_80;
```

13. **Add a column `full_name` to a backup table and populate it with first_name || ' ' || last_name for all existing rows (requires UPDATE after ADD; then you could add a default for new rows).**

```
ALTER TABLE EMP_BACK_80 ADD FULL_NAME VARCHAR(20) ;
UPDATE EMP_BACK_80 SET FULL_NAME=FIRST_NAME||','||LAST_NAME;
```

```
SELECT * FROM EMP_BACK_80;
```

4. **Create a table that has department_id, department_name, and a column `manager_name` (you would need to join hr.departments with hr.employees on manager_id to get manager's name).**

5. **. Create table `emp_job_salary` with columns job_id, min_sal, max_sal, avg_sal (use GROUP BY job_id with MIN, MAX, AVG on salary from hr.employees)**

```
CREATE TABLE EMP_JOB_SALARY AS
 SELECT JOB_ID,MIN(SALARY) AS MIN_SAL,MAX(SALARY)AS MAX_SAL,A
VG(SALARY) AVG_SAL
 FROM HR.EMPLOYEES
 GROUP BY JOB_ID

 SELECT * FROM EMP_JOB_SALARY
```

6. **Add a column that has a DEFAULT expression using SYSDATE and rename an existing column in the same table (two statements)**

```
ALTER TABLE EMP ADD D123 VARCHAR(20) DEFAULT SYSDATE;
SELECT * FROM EMP;
ALTER TABLE EMP RENAME COLUMN D123 TO D456
DESC EMP
```

7. **Create a table `emp_top_sal` with the same structure as hr.employees but only rows where salary is in the top 10 (use subquery: WHERE salary IN (SELECT ... ORDER BY salary DESC FETCH FIRST 10 ROWS ONLY)).**

```
create table emp_top_sal
as select *
```

```
 from(select * from emp
 order by sal desc)
  where rownum<=10;
```

8. **Create table `dept_emp_list` with department_id, department_name, and employee_count (count of employees per department).**

```
create table dept_emp_lis
    as select *
  from (select job,count(*)
  from emp
    group by job);
```

9. **Drop two columns from your backup table in one statement (if Oracle supports: ALTER TABLE … DROP (col1, col2)).**

```
alter table emp_top_sal drop (sal,comm);


select * from emp_top_sal;
```

10. **Create a table that contains only employees whose manager_id is not null and department_id is not null (all columns).**

```
create table dnam as select *
    from emp
    where empno is not null and
       comm is not null;

       select * from dnam;
```

11. **Add a column `salary_band` VARCHAR2(10) and update it with CASE (Low/Medium/High) based on salary; then add DEFAULT 'Medium' for new rows.**

```
alter table dnam add salary_band varchar(20) ;

update dnam
    set salary_band=sal;


SQL> select t.*,
    case when sal>1500 then 'high'
         when sal<1500 and sal>1200 then 'medium'
         when sal<1200 then 'low'
    else 'invalid'
    end
    from dnam t;

    alter table dnam add abc varchar(20) default 'medium';
```

12. **Create table `emp_duplicate_check` with employee_id, first_name, last_name, and a column `dup_count` showing how many employees share the same first_name and last_name (use analytic or self-join in CTAS).→self.join**

13. **. Create an empty table with the same structure as hr.employees and name it `emp_import_staging`.**

```
create table emp_import_staging as select * from emp where 1=
0;
select * from emp_import_staging
```

14. **Modify the data type of a column from NUMBER to VARCHAR2 (e.g. store employee_id as string). Oracle may require add new column, update, drop old, rename.**

```
alter table emp_import_staging modify deptno varchar(20)
desc emp_import_staging
```

15. **Create table dept_location_1700 from hr.departments where location_id = 1700.**

```
create table emp_deptno as select * from emp where deptno=10
select * from emp_deptno
```

16. **Add column `version` NUMBER DEFAULT 1 and `last_modified` DATE DEFAULT SYSDATE to backup table.**

```
alter table emp_deptno add( version number(10,2) default 1,
                            last_mft date default sysdate);

select * from emp_deptno
```

17. **Create table `emp_salary_range` with columns from hr.employees but only for salary between 5000 and 15000.**

```

create table emp_sala as select * from emp where sal >=500 an
d sal <15000

select * from emp_sala
```

18. **Truncate a table and then add a new column. Verify the table has 0 rows.**

```
select * from emp_sala
truncate table emp_sala
alter table emp_sala add de varchar(10)
```

19. **Create table `job_list` with distinct job_id from hr.employees and a literal column `category` with value 'HR'.**

```
select * from emp

create table job_list as select distinct job, 'hr' as categor
```

```
y from emp

select * from job_list
```

20. **Drop table `emp_structure` if it exists (Oracle: use PL/SQL EXECUTE IMMEDIATE 'DROP TABLE emp_structure'; with exception when table does not exist, or check user_tables first).**

```
select * from job_list

BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE job_list';
END;
```